PakEmis Sector Technical Note TN10

From Fuel Customers to Flight Geography: A Reproducible Aviation Emissions Account for Pakistan

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Abstract

Petroleum statistics classify aviation fuel by customer, whereas greenhouse-gas inventories classify flights by origin and destination. We develop a source-row-conserving bridge between these systems for Pakistan using annual fuel reports, five matched fiscal years of OECD aviation estimates and 42 monthly comparisons. In FY2023–24, commercial aviation fuel implies 2.032 Mt CO_2, 0.8% below OECD's combined domestic and international territory estimate. This aggregate agreement is misleading: resident-customer fuel is 28.6% above OECD's resident-carrier estimate, while foreign-customer fuel is 23.1% below its non-resident estimate, causing 97% of the component discrepancy to cancel.

The figure the national inventory carries is the aggregate movement model's 0.301 Mt CO_2e (0.299 Mt as CO_2 alone) — the estimator with the strongest activity evidence behind it, and the one every national product uses. Around it: the OECD resident-carrier partition gives 0.292 Mt of domestic CO_2, which is a calibrated consistency check rather than independent validation because OECD supplies that partition, and the route–aircraft Monte Carlo gives a mean of 0.244 Mt CO_2 with a 95% interval of 0.215–0.272 Mt. Pakistan's official 2022 reporting instead applies an 80/20 domestic–international allocation that is not consistent with the petroleum and flight-geography evidence examined here. The framework conserves every reported fuel row, keeps unresolved quantities explicit and separates calibration, external evaluation, structural uncertainty and parametric uncertainty. It provides a transferable method for reconciling purchaser-based petroleum statistics with flight-based aviation accounts.

The common annual core now reaches FY2024–25. Its LTO allocation uses the corresponding complete PCAA airport-movement source, with the reference aircraft composition explicitly carried where own-year fleet evidence is absent. The detailed route, OECD and structural comparisons in this note remain anchored to FY2023–24.

Keywords: civil aviation, fuel reconciliation, flight geography, international bunkers, structural uncertainty, IPCC 2006 Guidelines, Pakistan

PakEmis Sector Technical Note TN10. Technical documentation of one sector module of the PakEmis release identified below, deposited with the data as part of the documentation record. The national data-description article (`paper_national/`) is the only document that states the national totals as its own result.

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Release provenance. This document describes PakEmis under the period, configuration and source boundaries specified in its methods. The identifiers below link it to the accompanying products and source register.

Release version	`1.0.0rc69`
Commit	`d8d874622f9a`
Reporting period	July 2015–June 2025 (ten fiscal years) (2015-07-01 to 2025-06-30)
Reference	S_TAN (household survey; TAN nitrogen)
FY2023–24 total	444.61 Mt CO_2e (AR5) under S_TAN, the reference; 449.85 Mt under L_TAN
Release manifest	`ac0ffe93ad9fcd88` (SHA-256 of `outputs/release_manifest.json`, first 16)
Source manifest	`c1be6d314cfce1ea` (SHA-256 of `data/manifests/source_manifest.csv`, first 16)
Data-description article	`paper_national/` (ESSD, in preparation), same release

1. Introduction

National aviation accounts must distinguish three concepts: who bought the fuel, which carrier operated the aircraft and whether a flight stage was domestic or international. Petroleum

statistics commonly follow purchasers. IPCC inventories instead classify each flight from its origin and destination, irrespective of airline nationality (IPCC, 2006). A national airline can therefore buy fuel recorded under a domestic customer label while using much of it on international services.

Pakistan makes the consequences visible. A movement-based calculation placed domestic aviation near 0.30 Mt CO_2e in FY2023–24, but had no fuel control and carried a broad representation uncertainty. Public Oil Companies Advisory Council (OCAC) statistics report JP-1 separately for PIA, other Pakistani-airline customers and foreign aviation. Read literally, the resident-customer rows imply almost four times the movement estimate. That is not evidence of fourfold domestic emissions: "Domestic Avi–PIA" identifies a purchaser, not a domestic flight boundary.

The problem is a conserved partition rather than another uncertain percentage. Reassigning a tonne from domestic aviation also changes international bunkers or another aviation account. The source row must therefore remain intact while its destinations, method and unresolved quantity stay visible. A second distinction follows. OECD estimates monthly territory- and residence-based aviation emissions from flight data (Garsous et al., 2022). Those data can help partition resident-customer fuel, but an estimate produced with that partition cannot then be presented as an independent OECD validation. Gross fuel and flight accounts can still be compared more independently, and their resident and foreign components can reveal whether an apparently good total is internally coherent.

Aircraft-performance detail addresses a different layer. ICAO route factors and OpenAP synthetic profiles can test the consequences of representative distance, route mix and aircraft assignment (ICAO, 2024; Sun et al., 2020). Climate TRACE provides an additional airport-level comparison, although surveillance and performance inputs may overlap with other flight models (Climate TRACE, 2026). These sources strengthen the evidence hierarchy only when their roles and shared inputs are declared in advance.

We therefore ask one question: can purchaser-based aviation-fuel statistics be reconciled with flight-geography accounts without allowing aggregate agreement to conceal errors in the underlying components? We answer it with nine OCAC fuel years, five complete fiscal-year OECD comparisons, 42 matched months, Pakistan's official UNFCCC reporting and a hierarchy of domestic activity models.

The paper makes three contributions. First, it introduces a source-row-conserving bridge that separates purchaser, carrier residence and flight geography across domestic aviation, international bunkers, military/government, general aviation, other uses and an explicit residual. Second, it shows that near-closure of Pakistan's gross commercial aviation account conceals persistent resident and foreign discrepancies of opposite sign. Third, it evaluates domestic aviation through aggregate-movement, route-resolved and route–aircraft estimates, with calibration, external comparison, structural scenarios and parametric uncertainty kept distinct. Movement-to-cycle conversion, flight-geography classification and full-flight fuel remain executable accounting safeguards rather than headline contributions.

2. Data and Accounting Boundaries

Pakistan Civil Aviation Authority (PCAA) statistics provide the activity controls. The FY2023–24 airport report records 40,600 domestic and 95,267 international commercial movements, 21,223 non-commercial movements and 5.680 million domestic passengers (Pakistan Civil Aviation Authority, 2024b). A separate airline report gives departures, arrivals, passengers, cargo and mail by operator (Pakistan Civil Aviation Authority, 2024a). One movement is one take-off or landing; the national domestic activity control is therefore 20,300 flight stages. The historical movement series covers FY2006–07 to FY2023–24.

OCAC sectoral reports provide the fuel control. In FY2023–24 they record 193 813 t of JP-1 for "Domestic Avi–PIA", 166 119 t for "Domestic Avi–Others" and 283 147 t for "Overseas/Foreign Aviation", with government JP-1, JP-8 and 100LL retained separately (Oil Companies Advisory Council, 2024). Final reports supply nine annual fuel accounts from FY2016–17 to FY2024–25; public cumulative reports supply 42 months that overlap OECD completely. Revisions and category-label changes are versioned rather than silently harmonised (Oil Companies Advisory Council, 2026).

OECD monthly aviation estimates cover calendar 2019–2024 and distinguish domestic and international territory emissions from resident- and non-resident-carrier activity (Garsous et al., 2022). In FY2023–24 they report 235.8 kt CO_2 for domestic territory aviation and 1812.0 kt for international aviation. Resident carriers account for 227.0 kt of domestic and 657.3 kt of international-departure emissions. Their domestic share supplies the central fuel partition; OECD remains an external comparison for models that do not use that share.

Pakistan's Biennial Transparency Report, Third National Communication and 2026 Common Reporting Table (CRT) package provide the official national comparison (Government of Pakistan, 2024, 2025, 2026). The CRT workbook supplies calendar-2022 emissions, energy activity, fuel notation and an explicit domestic/international distribution. Climate TRACE v7 supplies national monthly and airport-source records for an exact FY2023–24 temporal and spatial comparison (Climate TRACE, 2026). ICAO route factors, OpenAP, public ADSB.lol traces and ERA5 meteorology support the model-resolution and sensitivity analyses; their detailed adapters and coverage limits are documented in the supplement (Sun et al., 2020; ADSB.lol, 2024; Copernicus Climate Change Service, 2026).

Table 1 fixes each source's evidentiary role before results are examined. In particular, "external" does not automatically mean independent, and a calibration input is never counted as validation.

Source/comparison	Role	Shared input or boundary constraint	Evidence class
PCAA airport and airline reports	Movement, passenger and margin controls	One reporting system for activity and load checks	Targeted control
OCAC total vs OECD combined	Gross fuel–flight closure	Carbon conversion only	Largely independent
OCAC bridge vs OECD domestic	Resident-customer partition	OECD share used directly	Calibration
Movement/route models vs OECD	Domestic territory comparison	Possible generic ICAO conventions	External, partial
Pakistan CRT vs OECD	Official-boundary diagnostic	Inventory year only	External, largely independent
PCAA airports vs Climate TRACE	Spatial ordering	Possible surveillance/performance overlap	Corroboration
ADSB.lol and ERA5	Coverage and weather sensitivities	Reconstructed routes or airport coordinates	Supplementary diagnostics

Table 1: Data roles and comparison independence. The evaluation does not promote any source to a universal truth benchmark.

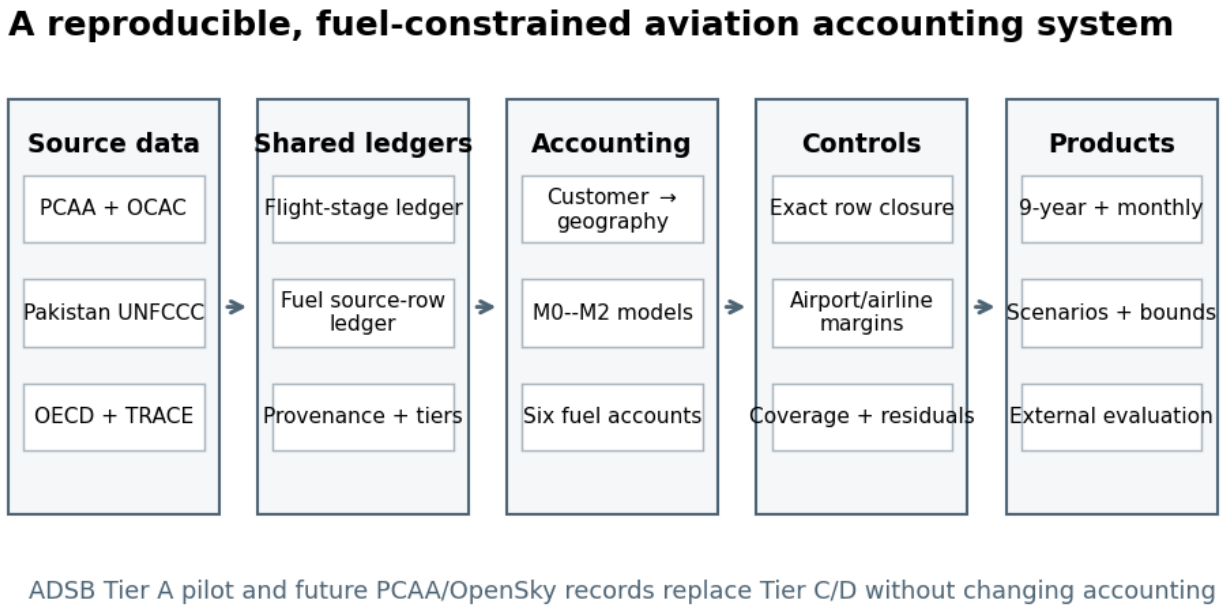


Figure 1: Data sources populate conserved fuel and flight ledgers, from which accounting products, domestic model estimates and external evaluations are generated.

3. Methods

3.1. Conserved customer-to-geography fuel ledger

Every OCAC record retains its period, customer category, fuel type, reported tonnes, source file and revision status. It is assigned across six non-overlapping destinations: IPCC domestic civil aviation (d), international bunkers (i), military/government 1A5b (m), general aviation (g), export/other (o) and unresolved residual (u). For each source row r,

$$F_r = F_{r,d} + F_{r,i} + F_{r,m} + F_{r,g} + F_{r,o} + F_{r,u}. \tag{1}$$

Closure is asserted before aggregation. Missing evidence increases $F_{r,u}$; it never triggers a national rescaling.

Three structural mappings test the customer-to-geography boundary. A literal mapping assigns resident-airline JP-1 to domestic aviation. A movement-share mapping divides it according to domestic and international flight stages. The central calibrated mapping uses the OECD resident-carrier share

$$s_d = \frac{E^{\mathrm{OECD}}_{\mathrm{resident,domestic}}}{E^{\mathrm{OECD}}_{\mathrm{resident,domestic}} + E^{\mathrm{OECD}}_{\mathrm{resident,intl.from}}}, \tag{2}$$

which is 25.674% in FY2023–24. Foreign-customer JP-1 is assigned principally to international bunkers; government aviation JP-1 and JP-8 are military/government candidates; 100LL is a general-aviation candidate; and government-other JP-1 remains unresolved. Alternative carrier-specific movement and modelled-fuel shares preserve separate PIA and other-airline networks while remaining constrained to the same aggregate OECD share. These mappings are reported as scenarios, not pooled into a confidence interval.

Three definitions are fixed as accounting invariants. PCAA movements are divided by two to obtain domestic stages because each flight generates a departure and an arrival. Domestic status requires both endpoints in Pakistan, irrespective of customer or carrier residence. Domestic greenhouse gases use full-flight fuel; international airport LTO emissions remain a separate local-air-quality product (IPCC, 2006).

3.2. Domestic activity and fuel models

The aggregate movement model (M0) multiplies domestic stages by a representative distance and year-varying fleet intensity:

$$E_y = N_y \, \bar{d}_y \, I_y, \tag{3}$$

where N_y is flight stages, $\bar{d}_y$ mean distance and I_y fuel and emission intensity. The route-resolved model (M1) replaces the representative stage with an annual origin– destination matrix constrained to PCAA airport margins, airline departures and routes documented for FY2023–24. Unsupported movements remain an unallocated residual. Minimum and maximum feasible distance are solved over the same route graph and control margins.

The route–aircraft model (M2) assigns probabilistic aircraft-family shares by operator and route and generates phase-resolved synthetic profiles with OpenAP (Sun et al., 2020). Operator passengers, cargo and mail constrain central payload; route detour, taxi time and aircraft assignment remain uncertain. Unsupported aircraft use a separately identified ICAO factor, while route, operator and endpoint residuals are completed through explicit lower-tier methods. A common-support comparison applies all three models to the same 17,491 stages before those residuals are added, preventing model resolution from being confused with coverage. Synthetic route strata are never labelled as observed flights.

The flight-stage ledger is source-neutral: observed records may replace reconstructed strata without changing accounting or emissions code. The detailed route optimiser, source tiers, OpenAP range checks, ADS-B endpoint rules, payload construction and coverage activation gate are provided in the supplement.

3.3. Evaluation and comparison design

OCAC cumulative monthly values are first-differenced and required to reproduce each final annual source row. OECD records are aggregated to identical July–June windows. For component j and month or year t,

$$R_{t,j} = \frac{F^{\mathrm{OCAC}}_{t,j}\, 3.16}{E^{\mathrm{OECD}}_{t,j}}. \tag{4}$$

We compare resident, foreign and combined commercial accounts and calculate the fraction of opposite component differences that cancels in their total:

$$C = 1 - \frac{|\Delta_{\mathrm{resident}} + \Delta_{\mathrm{foreign}}|}{|\Delta_{\mathrm{resident}}| + |\Delta_{\mathrm{foreign}}|}. \tag{5}$$

Five complete fiscal years support annual comparison and 42 matched months test whether annual closure is temporally stable.

The central fuel bridge uses OECD's resident share and therefore tests consistency, not independent accuracy. The strongest gross closure compares OCAC commercial JP-1 with OECD combined territory aviation. Aggregate movement, route-resolved and route–aircraft models are compared externally with OECD domestic territory emissions. Pakistan's CRT tests the official boundary implementation; Climate TRACE tests fiscal-year magnitude and airport ordering but may share surveillance or performance inputs. PCAA movement, passenger and route constraints target specific internal failure modes.

Public ADSB.lol and ERA5 experiments remain supporting diagnostics. The former tests whether a small public trace sample can replace reconstructed stages; the latter tests the sensitivity of representative OpenAP flight duration to route winds. Neither is expanded into an observed annual inventory or counted as fuel validation.

3.4. Parametric and structural uncertainty

Parametric and structural uncertainty are evaluated separately. The aggregate movement model retains its legacy activity/fleet representation term. The completed route–aircraft

model uses 20,000 flight-stage Monte Carlo draws within each structural configuration. Aircraft family is shared within operator–route strata, detour across reciprocal routes, taxi time by airport and carbon conversion globally. Payload is sampled around PCAA passenger, cargo and mail controls. CH4, N2O and net calorific value enter only the secondary CO_2e result.

Alternative feasible route matrices and unsupported-activity completion methods define a structural envelope; customer-to-geography mappings remain a separate accounting scenario. The route–aircraft model does not inherit the aggregate model's ±60% representation term. A drop-one variance screen identifies which named variables drive its interval. Automated QA/QC asserts source-row and monthly closure, route-bound ordering, common support, coverage completion, phase-fuel closure, movement balance and agreement between generated outputs and the manuscript. The complete uncertainty register and assertion catalogue are supplementary.

4. Results

4.1. Do the gross fuel and flight accounts agree?

OCAC commercial JP-1 corresponds to 1940.9 kt CO_2 in FY2022–23 and 2032.1 kt in FY2023–24. These values are respectively 0.16% and 0.77% below OECD's combined domestic and international territory estimates (Table 2). Agreement in the latest two years is therefore close at the gross commercial boundary.

The longer record prevents the two-year result from being treated as universal. Across the five complete OECD overlaps, the combined OCAC/OECD ratio is 0.908, 0.892, 0.935, 0.998 and 0.992 from FY2019–20 to FY2023–24. The OCAC reference-span series also captures the fall in commercial JP-1 from 730.7 kt in FY2017–18 to 328.7 kt in FY2020–21 and its subsequent recovery. Monthly combined MAPE is 7.39% in FY2022–23 and 6.92% in FY2023–24. Annual near-closure is thus reproducible in the latest years, but neither historically invariant nor the product of twelve identical monthly ratios. Full annual and monthly panels are reported in the supplement.

4.2. Does agreement survive component decomposition?

Here, "resident customers" are PIA and other Pakistani-airline fuel purchasers in OCAC, whereas "resident carriers" are Pakistani airlines identified in the OECD flight account. The foreign comparison follows the equivalent OCAC and OECD groups.

Aggregate agreement does not survive this decomposition. Resident-customer fuel is 29.9% and 28.6% above OECD resident-carrier emissions in FY2022–23 and FY2023–24, whereas foreign-customer fuel is 23.1% below OECD non-resident emissions in both years. The opposite differences cancel by 99.4% and 97.0%, leaving a small combined discrepancy (Figure 2). Monthly resident ratios average 1.301 and 1.288, while foreign ratios average 0.770 and 0.768. The sign of the mismatch therefore persists across both years and months.

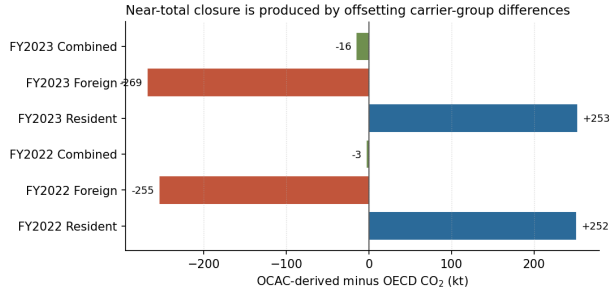


Figure 2: OCAC-derived minus OECD emissions by resident and foreign component. Aggregate agreement is produced by differences of opposite sign.

The conserved ledger shows where each reported tonne goes under the central FY2023–24 mapping (Table 3). PIA and other resident-customer JP-1 contribute 49,760 and 42,649 t to domestic aviation. Their remaining 267,523 t, together with the foreign-customer row, enters the international-bunker candidate. Government and 100LL streams retain separate destinations, and an explicit 12 282 t residual remains unresolved. Every row closes without using that residual to improve the apparent agreement.

Because customer classes cannot directly supply the domestic boundary, we next compare alternative estimates of domestic aviation.

4.3. What does the bridge imply for domestic aviation?

Using OECD's resident-carrier domestic share assigns 92,409 t of fuel, or 292.0 kt CO_2, to domestic aviation (Table 4). The aggregate movement estimate is 298.9 kt, only 2.3% higher. Because OECD supplies the partition, this agreement is a consistency check. Against the independent OECD domestic-territory value of 235.8 kt, the aggregate movement model is 26.8% higher, the route-resolved model is 14.8% higher and the completed route–aircraft model (M2) is 4.7% higher.

The same ordering broadly replicates in FY2022–23: aggregate movement fuel is 101,196 t, route-resolved fuel 90,446 t, the calibrated bridge 94,314 t and OECD domestic territory 237.2 kt CO_2. Including CH4 and N2O gives 0.294 Mt CO_2e for the FY2023–24 bridge, compared with 0.301 Mt for the aggregate movement model. The central bridge also assigns 550,670 t of commercial JP-1 to full-flight international bunkers, or 1.740 Mt CO_2. That quantity is distinct from the 210.4 kt international LTO estimate at Pakistani airports. The non-commercial fuel the bridge identifies is now priced too, as memo rows beside the bunkers: 120,758 t of military and government jet fuel (381.6 Gg CO_2), 4,542 t of 100LL general aviation (14.1 Gg) and 12,282 t of unresolved government JP-1 (38.8 Gg)—together about 0.434 Mt, larger than the reported domestic sector, outside the 1.A.3.a total because IPCC books military aviation under 1A5b and no record attributes the activity; the scope choice is priced against the national total as three independent configurations, one per fuel evidence class, and the same streams ship as a first-class OBSERVED-UNALLOCATED

Component	FY2022–23				FY2023–24			
	OCAC kt fuel	OCAC kt CO_2	OECD kt CO_2	Ratio	OCAC kt fuel	OCAC kt CO_2	OECD kt CO_2	Ratio
Resident carriers	346.0	1093.3	841.8	1.299	359.9	1137.4	884.3	1.286
Foreign carriers	268.2	847.6	1102.3	0.769	283.1	894.7	1163.5	0.769
Combined commercial	614.2	1940.9	1944.0	0.998	643.1	2032.1	2047.8	0.992

Table 2: Matched carrier-component comparison. Government fuel is excluded from the commercial boundary.

OCAC source row	Reported t	Domestic	Bunker	Military	General aviation	Residual
Domestic Avi–PIA, JP-1	193,813	49,760	144,053	—	—	—
Domestic Avi–Others, JP-1	166,119	42,649	123,470	—	—	—
Overseas/Foreign Aviation, JP-1	283,147	—	283,147	—	—	—
Government Aviation, JP-1	2,973	—	—	2,973	—	—
Government Other, JP-1	12,282	—	—	—	—	12,282
Government Other, JP-8	117,785	—	—	117,785	—	—
All reported 100LL	4,542	—	—	—	4,542	—
Total	**780,661**	**92,409**	**550,670**	**120,758**	**4,542**	**12,282**

Table 3: FY2023–24 source-row-conserving fuel bridge under the OECD-calibrated resident share. Dashes denote zero assignments.

Estimate	Fuel (t)	kt CO_2	Role
Aggregate movement (M0)	94,576	298.9	Unconstrained
Route-resolved (M1)	85,669	270.7	Unconstrained
Calibrated fuel bridge	92,409	292.0	OECD-calibrated
Movement-share bridge	107,555	339.9	Structural
Literal customer labels	359,932	1137.4	Diagnostic
Completed route–aircraft model (M2)	78,139	246.9	Provisional
OECD domestic territory	—	235.8	External

Table 4: FY2023–24 domestic CO_2 estimates. Different evidence roles preclude averaging the rows into one preferred value.

ACCOUNT (`outputs/aviation_unallocated_fuel_fy2023_24.csv`: each identified tonne with a closed evidence class, its disposition, and the record that would close it), so booked-outside-the-boundary can never read as absent, because military attribution, general-aviation activity and the bridge residual are separate questions. The military/government, general-aviation and unresolved-JP1 cases remain separate additions to the supported reference boundary. Their native emissions and updated national consequences are supplied in the configuration product; they are not absorbed into the central domestic estimate.

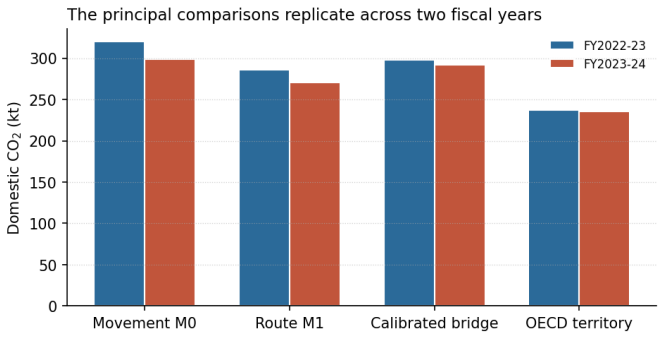


Figure 3: Two-year comparison of aggregate movement, route-resolved, calibrated-bridge and OECD domestic estimates.

Carrier-specific scenarios change attribution within the resident rows but not the calibrated total. After constraining separate PIA and other-carrier movement shares to the same aggregate, their domestic shares are 20.0% and 32.2%. Partial modelled-fuel shares are not used because only 27.9% of resident international departures are reconstructed.

4.4. Are the findings supported by other evidence?

Pakistan's official calendar-2022 CRT reports 654.1 kt domestic and 163.5 kt international aviation CO_2, compared with matched OECD values of 232.9 and 1713.3 kt (Figure 5a). The official workbook applies an explicit 80% domestic and 20% international fuel distribution, labels the activity aviation gasoline and marks jet kerosene not estimated. Its domestic and international ratios to OECD are therefore 2.808 and 0.095. The reported allocation is not consistent with the petroleum and flight-geography evidence examined here, and its institutional derivation could not be reconstructed. The supplement provides the workbook hash and cell-level audit.

Climate TRACE gives 255.9 kt domestic and 1950.9 kt international CO_2 for FY2023–24, 8.5% and 7.7% above OECD. Its airport ordering is strongly associated with PCAA activity: Spearman's $\rho = 0.971$ across 16 domestic airports and 0.948 across 11 international airports (Figure 5b). This supports the broad spatial pattern, not independent fuel-burn accuracy.

4.5. What remains uncertain?

The documented route graph allocates 20,098 of 20,300 domestic stages. Conditional on PCAA margins and supported links, its minimum, central and maximum mean distances are 485.2, 486.5 and 503.0 nmi. On identical 17,491-stage support, aggregate factors produce 81,490 t fuel, route-specific distance 76,102 t and route–aircraft profiles 73,439 t. Route resolution lowers fuel by 6.6% and the aircraft/performance layer by a further 3.5%. Explicit completion of unsupported aircraft and unallocated activity yields 78,139 t.

For the central structure, the route–aircraft Monte Carlo gives mean domestic emissions of 0.244 Mt CO_2 from 20,000

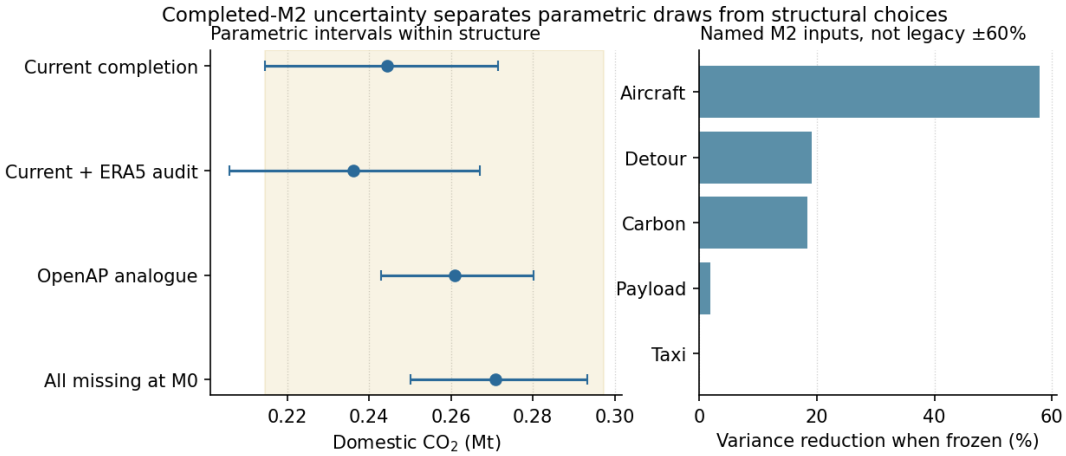


Figure 4: Route–aircraft Monte Carlo intervals, structural envelope and drop-one variance diagnostics for the completed route–aircraft model.

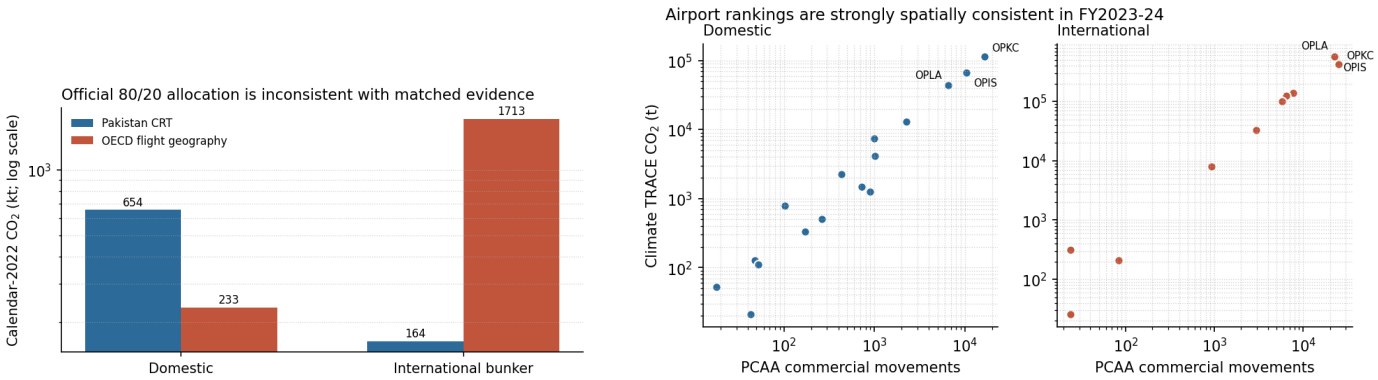


Figure 5: External diagnostics. (a) Pakistan's calendar-2022 CRT categories versus matched OECD flight-geography estimates. (b) FY2023–24 PCAA airport movements versus Climate TRACE airport CO_2.

draws (95% interval 0.215–0.272 Mt). Alternative route graphs and completion methods widen the endpoint envelope to 0.215–0.297 Mt. Aircraft assignment reduces simulated variance by 57.9% when frozen, followed by detour (19.2%) and carbon conversion (18.4%); these drop-one responses are non-additive. The remaining uncertainty is consequently tied to identifiable flight-stage quantities rather than the aggregate model's umbrella representation term.

Supplementary tests support the treatment of the completed route–aircraft model. A physical range assertion identified and corrected an OpenAP distance-unit mismatch, after which the common-support performance difference fell from 39.4% to 3.5%. Sampled ERA5 winds changed representative fuel by −3.9% on average. A four-day ADSB.lol pilot provided no usable Pakistan-airport stages despite passing synthetic and same-archive positive controls; reconstructed activity is therefore retained rather than expanded from inadequate observations.

5. Discussion

The central result is not simply that two national totals agree. In the latest two fiscal years, OCAC commercial fuel and OECD combined territory aviation differ by less than 1%, but resident and foreign components differ by about +29% and −23%. The signs persist in the monthly record, and the five-year combined ratio ranges from 0.892 to 0.998. Aggregate closure is therefore a useful check on the gross boundary, but it cannot establish that customer categories have been assigned correctly. Indeed, the closer the cancellation, the easier it is for a component error to disappear from the national total.

The conserved ledger changes how that evidence is interpreted. It prevents a customer label from being promoted directly to an IPCC category, preserves the international counterpart of every domestic assignment and leaves unsupported fuel as a statistical discrepancy. It also separates four quantities that serve different users: domestic aviation under the IPCC flight boundary, full-flight international bunkers, resident-carrier emissions and emissions located physically over Pakistan. A single fuel-sales label cannot represent all four.

Pakistan's official submission demonstrates the policy relevance. Its 80/20 distribution produces domestic and international estimates on opposite sides of the flight-geography evidence by factors of 2.81 and 0.095. The official values should not be dismissed merely because they differ from OECD, which is itself modelled. However, the aviation-gasoline label, the absence of estimated jet kerosene and the inability to reconstruct the allocation from public documentation mean that the split warrants institutional review. The appropriate next step is to reconcile the CRT activity data with OCAC jet-fuel rows and actual domestic and international flight stages, rather than replace one opaque percentage with another.

The hierarchy of domestic estimates localises the remaining scientific uncertainty. Across the complete inventory, replacing the 550-nmi representative stage with the constrained route matrix lowers fuel by 9.4%. On the identical 17,491-stage subset used for the resolution experiment, the corresponding reduction is 6.6%. Adding aircraft-specific OpenAP profiles on that common support lowers fuel by a further 3.5% after a physical range check corrected the earlier software interface error. The route–aircraft Monte Carlo mean of 0.244 Mt CO_2 is close to OECD's 0.236 Mt, but this is model-to-model agreement, not observed block-fuel validation. Its probabilistic and structural ranges identify aircraft assignment, detour and completion rules as actionable uncertainties. PCAA passenger, cargo and mail data already constrain payload; registrations, route-complete international activity and airline block fuel would test the remaining assumptions directly.

The external comparisons have deliberately different roles. Climate TRACE reproduces the airport ranking and broadly agrees on fiscal-year magnitude, but may share flight-surveillance and performance inputs with OECD or a future ADS-B implementation. The negative ADSB.lol pilot does not undermine the accounting result: positive controls show that the adapter works, while the Pakistan sample does not meet the coverage gate. Observed flight records would replace one input tier and test the present estimates; they are not required to make source-row conservation or component cancellation valid.

The approach is transferable wherever petroleum statistics follow purchasers while greenhouse- gas inventories follow flight geography. The essential requirements are modest: preserve each fuel source row, distinguish customer from operator and endpoints, disclose the role and shared inputs of every comparison, and report structural mappings separately from parametric uncertainty. More detailed trajectory or aircraft models then improve a defined layer of the account rather than silently changing its boundary.

6. Limitations

The central domestic fuel partition is calibrated with OECD's resident-carrier share. It is therefore not independently validated by its subsequent proximity to OECD or to the aggregate movement estimate. The route reconstruction covers domestic activity well but only 27.9% of resident international departures, so it cannot yet replace that share. OCAC customer rows also lack airport- and operator-specific uplift detail, and the meaning of government-other JP-1 remains unresolved.

The flight-stage layer is predominantly reconstructed. Route evidence and PCAA margins bound distance, while passengers, cargo and mail constrain payload, but aircraft registrations, flight-specific mass, detour, taxi time, weather and block fuel are not observed. The public ADSB.lol pilot covers only four days and should not be generalised to all ADS-B providers. The extended fuel ledger is not an observed annual route–aircraft inventory: detailed route and fleet reconstruction is anchored to FY2023–24. FY2024–25 airport movements are own-year activity, while its aircraft composition is explicitly carried from the reference era.

OECD and Climate TRACE are modelled products rather than direct fuel measurements, and their surveillance or performance inputs may overlap. The CRT is an official comparator, not ground truth; the public workbook does not reveal the institutional derivation of its 80/20 split. Monthly OCAC values are differences of cumulative reports, so revisions can shift apparent fuel between adjacent months even when annual closure is exact.

Finally, general aviation, training and military activity are not separated operationally. The spatial product is limited to airport LTO: no province, district or route–altitude cruise allocation is inferred from synthetic trajectories. The admitted LTO pollutant estimate uses representative engines and omits observed taxi times, APU, ground support, non-exhaust sources and near-airport measurements; it is not a complete airport micro-inventory.

7. Conclusion

Pakistan's commercial aviation fuel and OECD flight estimates nearly close in aggregate in the latest two fiscal years, but that agreement is produced by large component differences of opposite sign. In FY2023–24 resident-customer fuel is 28.6% above the resident-carrier estimate, foreign-customer fuel is 23.1% below the non-resident estimate and 97% of the discrepancy cancels. Pakistan's official 80/20 allocation produces a still larger domestic– international mismatch. National agreement is therefore a necessary gross check, not evidence that purchaser categories implement flight geography.

The national inventory carries the aggregate movement model's 0.301 Mt CO_2e as domestic aviation's figure. The central bridge assigns 0.292 Mt CO_2; aggregate-movement, route-resolved and completed route–aircraft estimates are 0.299, 0.271 and 0.247 Mt as CO_2; the route–aircraft Monte Carlo mean is 0.244 Mt with a 95% interval of 0.215–0.272 Mt. These estimates place domestic aviation in a common broad range while showing that customer partition, aircraft assignment and incomplete international coverage remain the most consequential uncertainties.

The general contribution is an accounting method rather than one assertedly final national number. Every fuel source row is conserved, residuals remain explicit, evidence roles are declared and structural mappings are kept outside parametric confidence intervals. Observed flight stages and detailed uplift records can now test and replace provisional inputs without changing those rules.

Code and data availability

The calculation engine, normalized activity data, source hashes and every output quoted here are version-controlled in the inventory repository (https://github.com/PakAirQuality/pakistan-emissions-sectors). The supplement gives the full reproduction sequence, source hashes, uncertainty register and assertion catalogue. The multi-gigabyte ADSB.lol cache is not redistributed under the ODbL; only aggregate pilot diagnostics are released. Manuscript values are checked against generated outputs.

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